

Ana Yupanqui Ponce De Leon

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PROFILE

Bachelor of Engineering (Honours) - Software student at UNSW, with *High Distinction* Honours Thesis in Human-Computer Interaction. Hands-on experience in developing intelligent, interactive systems across game development (*Unreal Engine 5*), full-stack mobile applications, and UX/UI design. Particular interest in *frontend development*, *creative technology*, and *health tech*.

Proven ability to collaborate in multidisciplinary teams, manage complex tasks, and deliver innovative solutions under tight deadlines.

EDUCATION

Bachelor of Engineering (Honours) – Software The University of New South Wales (UNSW) , Sydney Honours Thesis (Supervised by Dr Benjamin Tag) – <i>Complete</i> : • <i>Applying Latent Profile Analysis for User Clustering in Human-Computer Interaction</i>. • Investigating model-based clustering approaches to generate behaviourally meaningful user groupings for persona development and adaptive system design.	2022–2026 Course work complete.
Bachelor of Engineering (Honours) - Engineering Pathway The University of Adelaide , Adelaide • Transferred to UNSW after Year 1. • GPA 5.00 (Credit average for Engineering subjects)	2021–2022
Bachelor of Nursing (Year 1 completed) The University of Adelaide , Adelaide • GPA 6.00 (Distinction average).	2020–2021
SACE Certificate Sacred Heart College , Somerton Park	2017–2019

TECHNICAL EXPERIENCE - Software Projects

Interactive Survival Game – Unreal Engine 5 (AI, HCI, Immersive Systems)

Designed and developed a fully playable third-person survival game exploring human–companion interaction and noise-driven AI perception systems.

- Implemented multi-agent AI using Behaviour Trees and Unreal AI Perception (sight + hearing) to model reactive zombie agents driven by environmental stimuli.
- Designed a real-time stress modelling system for an AI companion (dog) based on proximity to threats using dynamic float scaling and state transitions.
- Developed command-based interaction mechanics (follow, stay, quiet) with stress-dependent duration scaling using interpolation functions.
- Integrated physics-based object interactions (projectile cans triggering AI hearing events via ReportNoiseEvent).
- Built dynamic UI elements (goal states, stress bar, command timer) driven by real-time system state.
- Applied immersive design principles including sound spatialisation, perception prioritisation (hearing > vision), and environmental storytelling.
- Packaged and deployed a Windows build for distribution.

“Roots and Shoots Foundation” – Figma Full Prototype

- Designed and prototyped a nonprofit charity website in **Figma**, collaborating in a 4-person team to apply the **user-centred design process**.
- Conducted **user research** (interviews, questionnaires, personas, requirements analysis) to inform functional and non-functional requirements.

- Implemented accessibility features (WCAG guidelines) and ran **formal usability tests**, iterating on the design to improve clarity and inclusivity for diverse users.

“GreenGotchi” – Sustainability-Focused Mobile App – Figma, Typescript, React Native, Python APIs.

- Collaborated in a cross-functional Agile team through three sprints to design and implement a **gamified mobile app** encouraging sustainable travel.
- Developed and integrated **five APIs** using NSW Transport data, enabling users to earn points for eco-friendly travel choices and care for virtual “tree pets.”
- Contributed to **front-end design in React Native** and **backend microservices**, overcoming challenges with API integration, environment setup, and CI/CD pipelines.
- Delivered a functional MVP, pitched to stakeholders in a **Shark Tank-style presentation**, and refined the product using usability testing and feedback.

Deep Learning Sea Turtle Image Segmentation Project- Python, PyTorch, OpenCV

Developed and evaluated deep learning pipelines for semantic and instance segmentation across 8,700+ images in a group setting.

- Implemented and compared **kNN, DeepLabV3+, Mask R-CNN, and YOLOv8-Seg** architectures.
- Achieved state-of-the-art performance (mAP@50 = 0.97; IoU = 0.972).
- Addressed class imbalance and signal noise using augmentation, preprocessing, and hyperparameter tuning.
- Conducted structured experimental comparisons across architectures.
- Optimised performance under hardware constraints.

Satellite Communication Simulation – Java

- Designed and implemented a medium-scale simulation of satellite and device communication around Jupiter using **Object-Oriented Programming** principles.
- Developed robust code with **programming by contract**, ensuring correctness through testing and iterative refactoring.

Dungeonmania Game Backend Refactoring – Java

- Inherited and refactored a large monolithic codebase for a dungeon-style game, improving design quality by resolving code smells and applying design patterns.
- Extended functionality to support new requirements, implementing iterative design and evolution of features.
- Collaborated with a partner to navigate an unfamiliar system, demonstrating adaptability and effective teamwork.

TECHNICAL SKILLS

- **Programming & ML:** Python (PyTorch, OpenCV), Java, JavaScript, React, SQL, C.
- **Immersive & Interactive Systems:** Unreal Engine 5, Unity, Behavioural Trees, AI Perception
- **Design & Tools:** Figma, Canva, Adobe XD, Microsoft Office Suite, Git., Confluence, Jira
- **Development Practices & Research:** Agile methodologies, version control, debugging, usability testing, and user research.
- **Certifications:** Basic Life Support, Safe Use of PPE, Infection Prevention and Control.

RECENT WORK EXPERIENCE

Phlebotomist – Clinpath Pathology (part of Sonic Healthcare Limited)

2025 – ongoing

- Performed precise clinical procedures while prioritizing and maintaining patient safety and comfort.
- Adherent to strict clinical protocols and maintained accurate patient records.
- Worked flexibly across multiple locations, adapting to varying operational demands.
- Built strong communication and teamwork skills with high attention to detail in high-pressure environments.

ADDITIONAL INTERESTS

- **Languages:** Spanish & English (native), French (conversational).
- Dog training at the German Shepherd Club, SA.